

Bipul Banjade

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Professional Summary

AI & Systems Engineer with a high learning velocity, passionate about tackling ambiguous problems in AI and infrastructure. Experienced in building/optimizing AI/ML pipelines and developing full-stack AI enterprise solutions with significant impact.

Education

New York University

August 2022 - May 2026

Bachelor's, Computer Science

- Double Major: Physics. Relevant Coursework: Artificial Intelligence (Grade: A), Operating Systems, Data Structures & Algorithms (Grade: A), Computer Vision, Quantum Mechanics (Grade: A)

Professional Experience

Fanatics

New York, NY, USA

AI Software Engineer Intern

August 2025 - Present

- Developed a full-stack AI corporate invoice analyzer with a scalable FastAPI, Docker, and Redis backend and leveraged AWS Textract, GPT-4, and pgvector embeddings for automation, saving 70+ hours weekly across 20+ employees.
- Increased M&A and finance team productivity by 40-60% by integrating agentic AI configurations (Copilot, GPT, Zapier) to automate critical workflow processes such as data parsing and schema mapping, policy adherence and risk analysis.

Scale AI

Remote

Generative AI Engineer

April 2024 - July 2025

- Optimized FastAPI services with async endpoints, job handlers, and optimized PostgreSQL (indexes + prepared statements), reducing GenAI LLM training pipeline downtime by 60% and boosting task throughput 30%.
- Transformed raw sales data into schema-aligned outputs for leading companies by tuning agentic ML systems (time-series, regression), enabling critical business insights on demand forecasting and growth with robust data handling.
- Authored 250+ unique programs (Python, JS, C++) to generate high-quality training data for Tier-1 LLMs, ensuring robustness by rigorously addressing edge cases and algorithmic correctness.

New York University

New York, NY, USA

IT Research Technician - Undergraduate Research

September 2023 - December 2024

- Resolved technical issues for 300 + Bobst Library patrons daily, ensuring uninterrupted research access.
- Developed and deployed MacOS scripts, managing access control and reducing manual labor by 5+ hours/week.

Projects

CNN_Pytorch

New York, NY, USA

CV+ML - <https://github.com/BipulB7/CNN-Pytorch-Project>

September 2025 - October 2025

- Engineered a Convolutional Neural Network from scratch in PyTorch, manually implementing forward and backward propagation for all core layers (Conv2D, MaxPool2D, Linear) and achieving 96.44% test accuracy on the MNIST dataset.
- Executed a transfer learning strategy by fine-tuning a pre-trained ResNet-18 model for the 62-class EMNIST dataset, implementing an efficient phased training approach (linear probing, partial unfreezing) to adapt the model to a new task.

AI-Based Research Hub "COSMICO"

Remote

Full-Stack - <https://cosmicoai.netlify.app/>

February 2025 - May 2025

- Developed an end-to-end full-stack RAG AI research website using React/JS, Tailwind, Node.js, Express, and MongoDB, using cross-API logic amongst OpenAI and arXiv APIs for real-time paper retrieval, citations, and automated summarization.
- Implemented cross-API orchestration and cost-aware model selection (route to smaller vs. larger LLMs by task scope) yielding 60% API cost reduction and ~40% latency improvement (Turbo 3.5 vs. 4o).

ERC-20 Wallet Balance Service

New York, NY, USA

Backend (Py, Docker, FastAPI, Postgres, Redis) <https://github.com/BipulB7/ERC-20-Wallet-Service>

July 2025 - August 2025

- Built a FastAPI + PostgreSQL service to compute ERC-20 wallet balances at arbitrary timestamps using token transfers, price joins, and indexed caching, with idempotent backfills, reorg handling, and LRU "as-of" price lookups.
- Containerized full stack (API, DB, Redis) with Docker Compose for reproducible local/cloud deployment.

Skills

- **Languages:** Python, JavaScript, C/C++, Java, TypeScript, SQL
- **AI/ML:** RAG, PyTorch, VectorDBs, TensorFlow, Scikit-learn, LLM Integration, Langchain, Transformers, Keras
- **Frameworks:** FastAPI, React.js, Node.js, Express.js, NumPy, Pandas, Matplotlib
- **Databases/Infrastructures:** Postgres, MongoDB, Redis, Docker, REST APIs, Git, Embeddings